

YASHKUL', Russian FederationWMO: **348660**Lat: **46.183N**Lon: **45.350E**Elev: **-22**StdP: **14.71**Time Zone: **3.00 (E03)**Period: **94-15**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.1	1.9	-10.3	3.2	-4.7	-3.0	4.7	3.3	26.1	32.2	21.7	27.9	4.9	0	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	100.1	70.2	97.0	69.5	93.9	68.5	73.9	92.3	72.2	90.2	70.6	88.2	8.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.4	104.3	81.1	66.4	97.2	79.6	64.7	91.5	77.7	37.4	92.3	35.8	90.1	34.4	88.4	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.4	18.7	16.0		-9.7	104.5	10.7	4.3	-17.5	107.6	-23.7	110.2	-29.7	112.6	-37.5	115.7	
				-10.2	76.0	10.5	2.9	-17.8	78.1	-23.9	79.8	-29.9	81.4	-37.5	83.6	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	26.0	26.9	37.8	52.4	65.3	75.1	79.6	77.5	65.0	52.0	38.3	30.6	
	DBStd	21.01	11.57	11.76	7.67	7.59	7.84	5.97	5.28	6.68	7.47	8.50	9.40	9.67	
	HDD50	2869	746	645	380	58	3	0	0	0	2	76	358	601	
	HDD65	5963	1210	1065	842	382	95	4	0	2	89	408	800	1066	
	CDD50	3733	0	0	3	130	477	754	917	853	452	139	8	0	
	CDD65	1354	0	0	0	4	104	309	452	390	89	6	0	0	
	CDH74	16696	0	0	0	71	1234	3668	5698	4993	955	77	0	0	
	CDH80	8378	0	0	0	7	463	1792	3102	2696	305	13	0	0	
Wind	WSAvg	6.3	6.1	6.9	7.7	6.9	6.5	6.2	5.7	5.7	6.0	5.8	5.9	6.3	
Precipitation	PrecAvg	11.5	0.6	0.5	0.8	1.0	1.5	1.5	1.0	0.8	1.0	1.0	0.8	0.7	
	PrecMax	15.5	1.2	0.9	2.1	3.7	3.6	5.8	4.8	3.2	3.9	3.7	1.9	1.6	
	PrecMin	7.5	0.0	0.1	0.2	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.2	0.0	
	PrecStd	2.4	0.3	0.2	0.5	0.9	1.1	1.2	1.0	0.7	1.0	1.0	0.5	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.1	50.8	63.0	80.9	94.8	100.9	104.5	104.1	93.0	81.5	62.2	51.2	
		MCWB	43.7	44.4	52.7	61.5	68.4	70.1	69.5	71.1	69.6	62.0	54.5	47.7	
	2%	DB	44.0	45.5	57.9	75.7	89.2	97.0	100.2	100.2	87.1	75.5	57.8	47.5	
		MCWB	40.9	40.7	48.0	58.9	63.0	69.7	69.7	71.0	66.4	60.5	52.5	44.7	
	5%	DB	40.9	42.1	53.9	71.2	85.5	93.8	97.0	96.6	83.4	70.0	53.8	44.4	
		MCWB	38.6	38.6	45.7	57.4	62.9	68.1	69.1	69.7	64.7	58.2	49.6	42.0	
	10%	DB	38.4	39.5	49.8	67.1	81.5	90.0	93.7	93.1	79.7	65.0	50.7	41.5	
		MCWB	36.5	37.0	43.4	55.2	62.3	67.3	68.1	68.7	63.1	56.3	47.6	39.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.5	45.2	53.2	64.5	72.2	75.3	75.0	77.1	72.8	64.5	56.0	48.5
			MCDB	48.0	49.6	61.5	75.8	91.8	93.5	96.0	96.5	85.9	75.5	60.7	50.6
2%		WB	41.2	41.3	48.9	61.0	66.8	72.8	72.8	74.5	69.5	61.7	53.0	45.1	
		MCDB	43.5	44.6	55.8	72.3	82.3	90.0	93.9	92.4	82.4	70.8	56.9	47.2	
5%		WB	38.8	39.1	46.4	58.4	65.0	70.7	71.1	72.3	67.0	59.5	50.4	42.4	
		MCDB	40.6	41.5	52.7	69.2	79.9	87.4	90.7	89.5	79.1	68.3	53.4	44.0	
10%		WB	36.6	37.2	43.9	56.1	63.4	68.8	69.8	70.6	64.8	57.3	47.8	39.9	
		MCDB	38.2	39.2	49.4	65.6	77.2	84.9	87.9	88.2	76.1	64.2	50.5	41.5	
Mean Daily Temperature Range	5% DB	MDBR	10.8	12.5	16.5	20.6	22.2	23.1	23.2	24.4	22.8	18.2	13.0	9.7	
		MCDBR	11.6	14.3	23.1	26.7	29.2	29.1	28.3	29.9	27.6	25.2	16.9	12.1	
		MCWBR	8.7	10.4	14.2	12.5	10.3	9.8	8.6	10.9	11.4	11.9	11.7	9.7	
	5% WB	MCDBR	10.7	12.9	21.3	24.2	25.1	26.1	24.8	26.4	24.8	22.0	15.4	11.8	
		MCWBR	8.4	9.8	13.9	12.2	10.7	10.8	9.6	11.7	11.5	11.6	11.1	9.6	
Clear-Sky Solar Irradiance	taub	0.356	0.388	0.419	0.475	0.447	0.439	0.451	0.457	0.433	0.393	0.366	0.365		
	taud	2.167	2.098	2.085	1.975	2.137	2.214	2.190	2.151	2.162	2.235	2.224	2.154		
	Ebn at Noon	222	239	252	248	261	264	259	251	245	236	217	202		
	Edh at Noon	29	37	44	54	47	44	44	44	40	32	27	26		
All-Sky Solar Radiation	RadAvg	326	592	993	1519	1914	2114	2061	1837	1363	831	405	255		
	RadStd	72	105	115	131	150	116	111	105	101	101	62	32		